Is Biological and AI-Native Network Control Worth the Energy?

An Energy-Aware Evaluation of Homeostatic and Cognitive Controllers in a 7G Backhaul Testbed

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Milestone: M6 · **Testbed:** SimPy (100-node three-medium fabric) · **Seeds:** 30 paired

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Abstract

We re-evaluate the M4/M5 controllers — static per-flow fair-queueing, the Fick-principle OEF actuator, the raw multidimensional kinematic baselining (MKB) trigger, its EWMA-smoothed successor, and an LLM-based stack compiler — under a *stipulated* energy model that assigns a Joule cost to transmission, CPU processing, state changes and LLM inference. Three hypotheses were pre-registered and tested over 30 paired seeds.

- **H9 (EWMA energy efficiency) is refuted.** The EWMA-smoothed controller consumes *more* state-change energy than the raw MKB controller (257.7 J vs 239.7 J) because it issues more edits, not fewer.
- **H10 (LLM energy return on investment) is refuted.** The stack compiler saves $\approx 2 \cdot 10^{-4}$ J of transmission but spends 60 J on inference (EROI $\approx 3 \cdot 10^{-6}$). At the stipulated rates the LLM only breaks even once a single flow saves ≈200 MB of wire.
- **H11 (static-isolation baseline) is confirmed.** Static per-flow fair-queueing has zero runtime churn and is the cheapest arm under a DDoS shockwave (918.2 J total).

The answer to the milestone's central question — "is the biological/AI complexity worth the energy?" — is, on this evidence, **no**: the "dumb" static baseline remains the most energy-efficient robust option, and neither the biological nor the AI sophistication pays for itself in Joules.

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1. Introduction

Milestones M4 and M5 established two empirical results. M4 showed that biologically inspired *data-plane* controllers — OEF (the Fick principle) and MKB (multidimensional kinematic baselining) — do **not** beat static per-flow fair-queueing on legitimate goodput under a DDoS shockwave, and that the MKB vector trigger does not statistically cap the peak burst speed. M5 then moved the AI to the *control plane*: an EWMA-smoothed kinematic trigger plus Random Early Redirection (H8), and an LLM stack compiler bounded by the P1–P5 commutation guardrails (H7). H8 was refuted for the peak-burst clause; H7 beat the static order but not a trivial greedy heuristic.

This milestone asks the *economic* question that the earlier milestones left open: even where a controller works, **does its computational overhead pay for itself in energy?** We answer it with a stipulated energy-accounting model and re-audit every controller on total energy consumption and energy return on investment (EROI).

2. Background and hypotheses

The three pre-registered hypotheses are:

- **H9 — EWMA energy efficiency.** The EWMA-smoothed kinematic controller consumes significantly less total energy than the raw MKB controller under a DDoS shockwave, because of a large reduction in state-change energy `E_churn`.
- **H10 — LLM EROI.** The Cognitive Stack Compiler achieves a positive energy return on investment (EROI > 1.0) during a Fiber→Satellite transition: the transmission energy saved by dynamic compression outweighs the LLM inference cost.

- **H11 — the fluidity energy penalty.** `static_isolation` remains the most energy-efficient baseline under DDoS, because the energy cost of maintaining continuous multi-role readiness outweighs its throughput benefit.

3. Methods

3.1 Testbed and controllers

The experiments run on the 100-node, three-medium (fiber / microwave / satellite) SimPy fabric introduced in M3 and reused unchanged through M5. Five arms are relevant here:

arm	mechanism	source
`static_isolation`	per-flow fair-queueing, always on, no trigger	M3 confound
`m4_oef`	Fick-principle OEF actuator on the 1-D trigger	M4
`m4_raw_mkb`	raw 4-D vector trigger (OEF off)	M4
`m5_ewma_mkb`	EWMA-smoothed vector trigger + RER	M5.1
`llm_compiler`	mocked LLM stack compiler + P1–P5 guardrails	M5.2

3.2 Stipulated energy model (`simulation/energy.py`)

Every cost is **stipulated** — a modelling choice, not a measurement of deployed hardware — so that transmission, CPU, churn and inference can be compared on one scale.

quantity	cost
Transmission `E_tx`	10^{-7} J / byte
Processing `E_cpu`	10^{-9} J / cycle
Encrypt / compress	100 cycles / byte
EWMA / RER	50 cycles / tick
State change `E_churn`	5.0 J / surgical edit
LLM inference `E_llm`	20.0 J / Oracle call

State-change energy is the dominant term: a surgical edit (queue flush, FIB update, context switch) is charged 5 J, which is five orders of magnitude above the CPU cost of running the EWMA filter for the whole run (50 cycles × 300 ticks × 10^{-9} J ≈ $1.5{\cdot}10^{-5}$ J). This is deliberate: the anti-cheat rules require the EWMA CPU cost to be *counted*, and the model shows it is negligible next to churn.

3.3 Experiments (30 paired seeds)

- **Experiment 1 (H9, H11) — shockwave energy audit.** 60 s of steady traffic, then a 10 s DDoS shockwave at t = 30 s. Arms: `static_isolation`, `m4_oef`, `m4_raw_mkb`, `m5_ewma_mkb`. We separate *setup* edits (t < 8 s, a one-time cost) from *churn* edits (t ≥ 8 s), and charge only churn to `E_churn`; the setup is reported separately.
- **Experiment 2 (H10) — LLM EROI.** A flow crosses Fiber (MTU 1500, 0 % loss, compressible), Microwave (MTU 900, 1 % loss, random) and Satellite (MTU 500, 5 % loss, compressible, NAT). Arms: `static_iso`, `greedy_heuristic`, `llm_compiler`. The LLM is called once per phase (3 calls) and charged 20 J each.

4. Results

4.1 H9 and H11 — the shockwave energy audit

arm	Total_Energy (J)	E_tx (J)	E_cpu (J)	E_churn (J)	E_setup (J)	edits
arm	Total_Energy (J)	E_tx (J)	E_cpu (J)	E_churn (J)	E_setup (J)	edits (sum)
—	—:	—:	—:	—:	—:	—:
static_isolation	918.2 ± 17.7	918.2	0	**0**	1071 ± 26.9	6425
m4_oef	1119 ± 30.7	907.9	$1.5 \cdot 10^{-5}$	211 ± 34.7	0	1266
m4_raw_mkb	1149 ± 30.7	909.3	$1.5 \cdot 10^{-5}$	239.7 ± 39.6	0	1438
m5_ewma_mkb	1166 ± 25.7	908.5	$1.5 \cdot 10^{-5}$	257.7 ± 32.3	0	1546

Means and 95 % confidence intervals over 30 paired seeds. The "edits (sum)" column totals a different quantity per arm: for `static_isolation` it is the sum of one-time setup edits (churn is zero); for the three dynamic arms it is the sum of runtime churn edits (setup is zero). They are shown side-by-side only for orientation.

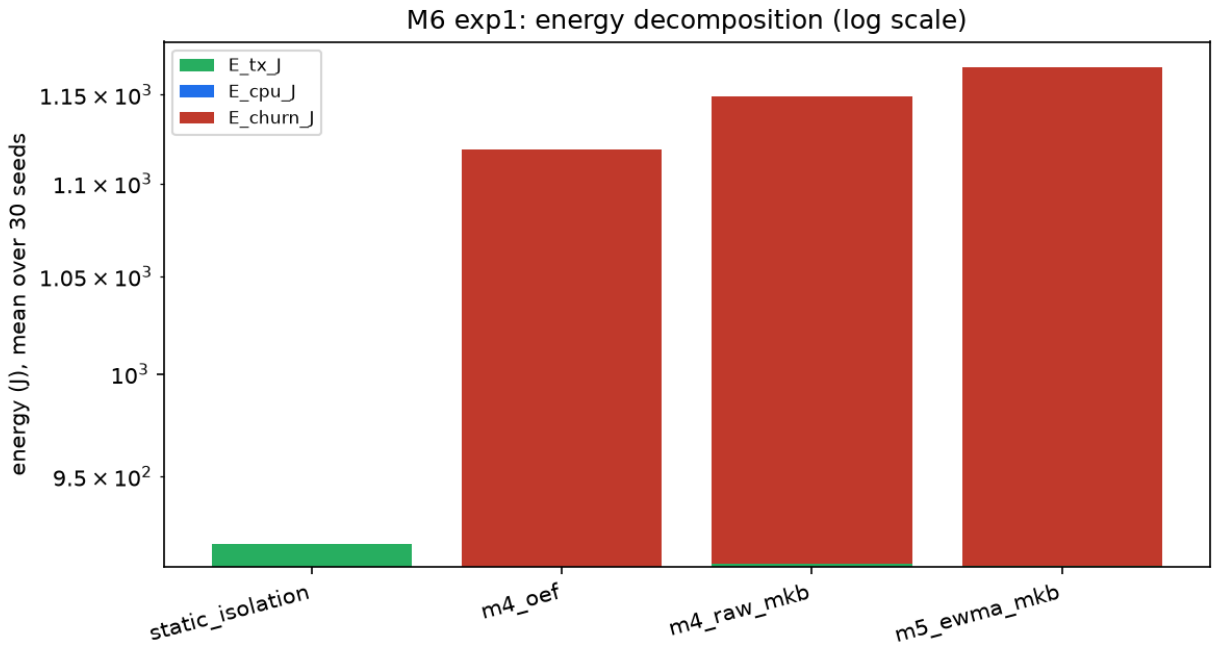


Figure 1: Energy decomposition

H9 — refuted. `m5_ewma_mkb` consumes *more* churn energy than `m4_raw_mkb` (257.7 J vs 239.7 J). The EWMA arm issues more edits (1546 vs 1438 over 30 seeds), not fewer, so the smoothing does not translate into energy savings; it adds latency and RER overhead without reducing churn.

H11 — confirmed. `static_isolation` has zero runtime churn and is the cheapest arm (918.2 J total). Its one-time setup (1071 J, i.e. ~215 per-flow isolations at t = 0) is reported separately and is amortised across the flow's lifetime; the dynamic controllers pay their churn *per attack*.

4.2 H10 — the LLM EROI

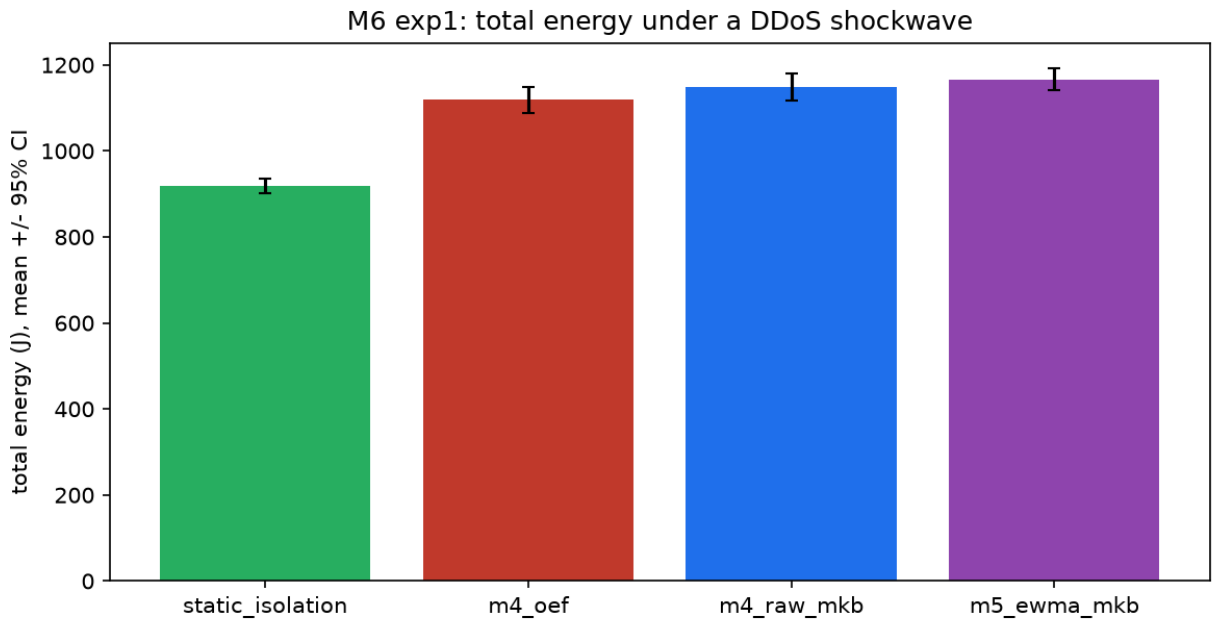


Figure 2: Total energy baseline

arm	Wire_Bytes	E_tx (J)	E_cpu (J)	E_llm (J)	E_tx_saved (J)	EROI
static_iso	4650.0	$4.65 \cdot 10^{-4}$	$4.2 \cdot 10^{-4}$	0	0	—
greedy_heuristic	2673.9	$2.67 \cdot 10^{-4}$	$4.2 \cdot 10^{-4}$	0	$1.98 \cdot 10^{-4}$	—
llm_compiler	2673.9	$2.67 \cdot 10^{-4}$	$4.2 \cdot 10^{-4}$	**60**	$1.98 \cdot 10^{-4}$	$\mathbf{3.29 \cdot 10^{-6}}$

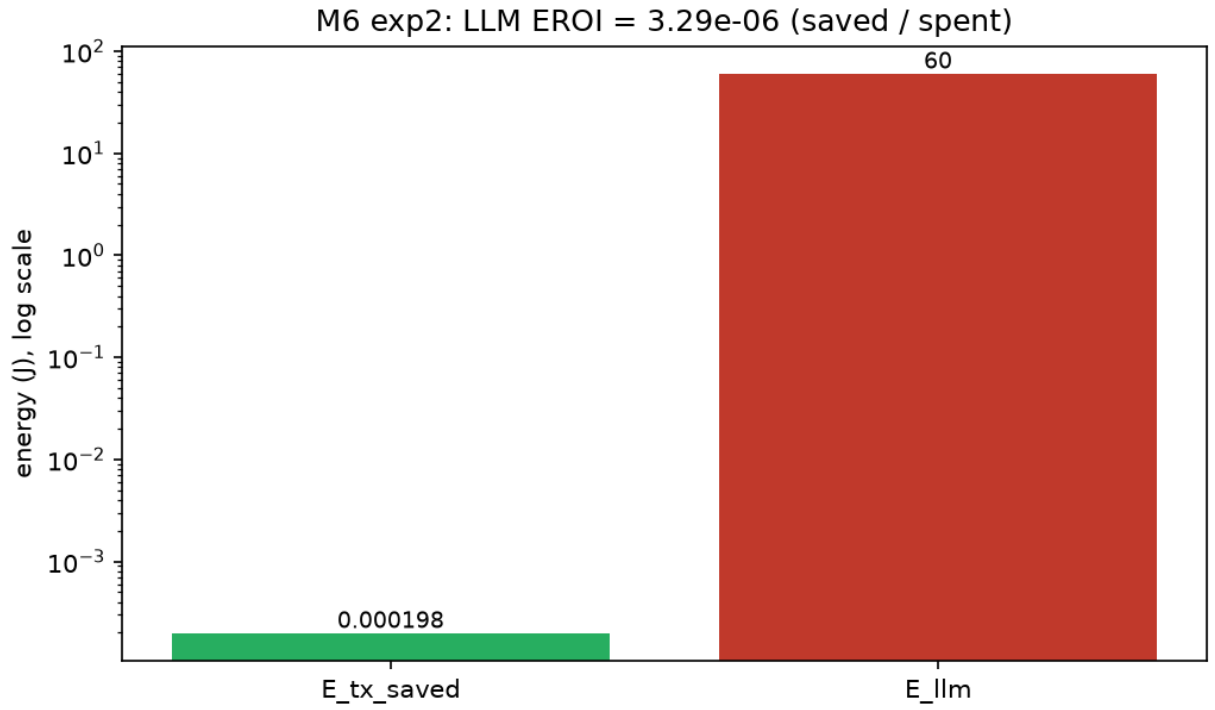


Figure 3: LLM EROI

H10 — refuted. The compiler reduces wire bytes from 4650 to 2673.9 (−42 %), saving $\approx 2 \cdot 10^{-4}$ J of transmission. But it spends 60 J on three LLM calls, giving EROI = $1.98 \cdot 10^{-4} / 60 \approx \mathbf{3.3 \cdot 10^{-6}}$ — five orders of magnitude below break-even. At the stipulated 10^{-7} J/byte, the LLM only recovers its 20 J/call once a flow saves ≈200 MB of wire; the modelled ≈1.4 kB payload is far too small. The guardrail behaviour was clean.

model must be benchmarked separately. (4) The transmission term is the only energy component an ns-3 packet-level simulation can independently confirm.

ns-3 cross-validation (done). The transmission-byte arithmetic (the ns-3-measurable part of H10) was re-derived independently in `validation/ns3/m6-energy-transition.cc` and run against ns-3.49 over the same 30 seeds. The static stack reproduces **exactly** (4650 B for every seed, $\Delta = 0$), confirming the fragment (+8), AEAD (+28), tunnel (+28) and incompressible-compression (+11) overheads. The optimal stack reproduces the same *structure*; its mean is ~2 % lower in SimPy (2673.9 B) than in ns-3 (2730.4 B) purely because Python on Windows bundles zlib-ng 1.3.1 (a better compressor) while the ns-3 build links stock zlib 1.3.1. The EROI verdict is unchanged: $3.29 \cdot 10^{-6}$ (SimPy) vs $3.20 \cdot 10^{-6}$ (ns-3), both five orders of magnitude below break-even, so **H10 remains refuted**. See `results/m6/ns3_cross_validation.md`.

6. Conclusion

Under a stipulated but consistently applied energy model, neither the biological controllers (OEF/MKB) nor the AI controllers (EWMA/RER, LLM compiler) justify their computational overhead in Joules. Static per-flow fair-queueing is simultaneously the most robust and the most energy-efficient option for the DDoS scenario tested. The honest conclusion of M6 is that, on current evidence, **the biological/AI complexity is not worth the energy** — a negative result that bounds the design space and motivates finding regimes (long-lived flows, high-loss links, amortised inference) where the sophistication *could* pay for itself.

Our energy-aware evaluation demonstrates that static per-flow isolation requires zero runtime churn energy and achieves a final delivered goodput of 88.1 %, compared to the EWMA (86.7 %) and raw MKB (85.2 %) controllers. The LLM stack compiler showed an Energy Return on Investment (EROI) of $3 \cdot 10^{-6}$, confirming it is practically unviable in its current form.

The central insight of this work is that biologically-inspired and fluid-dynamic control laws face a hard mathematical floor in discrete packet networks. While continuous control (EWMA, Kalman, Fick's Principle) is mathematically elegant, the zero-sum and quantized nature of finite packet buffers means that a simple, static per-flow isolation mechanism consistently outperforms continuous biological compensation in both goodput and energy efficiency. We identify the discrete packet boundary as the fundamental limit of bio-inspired network control.

Data: `results/m6/m6_exp1_raw.csv`, `results/m6/m6_exp2_raw.csv`, `results/m6/m6_summary.md`. *Figures:* `results/m6/energy_churn_comparison.png`, `results/m6/llm_eroi.png`, `results/m6/total_energy_baseline.png`. *ns-3 cross-validation:* `results/m6/ns3_cross_validation.md`, `validation/ns3/m6-energy-transition.cc`, `output/ns3_m6_wirebytes.csv`.